

Lakshyaraj Singh Rao

Backend Engineer · AI Infrastructure · Full-Stack · DevOps

Jaipur, India · Remote
raolakshyaraj@gmail.com
lakshyaraj-dev.vercel.app · GitHub · LinkedIn · npm

SUMMARY

Backend-focused engineer building production AI infrastructure. Four published npm packages (@ykstorg/goldset, quickdraw, tripwire, anvil — the last with npm build provenance) and a public GitHub Marketplace Action. Patterns from a solo production AI deployment, extracted into seven open-source repos spanning RAG, LLM safety, eval, benchmarking, Kubernetes/ GitOps, and webhook reliability.

TECHNICAL SKILLS

Languages	Backend / Data	Infra / AI
TypeScript · JavaScript Python · SQL Rust (learning)	Node.js · Express · Next.js PostgreSQL · pgvector Prisma 7 · Redis · BullMQ	Docker · Kubernetes · Helm Terraform · GitHub Actions · Vercel OpenAI · Anthropic · Ollama

OPEN-SOURCE PROJECTS

Anvil [Creator](#) · [github.com/ykstorm/anvil](#)

- Idempotent webhook → BullMQ worker pipeline: HMAC-SHA256 constant-time verify, dedupe by sha256(signature + payload), retry backoff [1s, 5s, 30s, 5m], dead-letter replay via a separate consumer.
- Five design contracts proven by tests in CI (Redis-backed). Ships a Terraform module + Helm chart. Published as \u{40}ykstorg/anvil v0.1.0 on npm with build provenance.

TypeScript BullMQ Redis Express Terraform Helm

Goldset [Creator](#) · [Maintainer](#) · [github.com/ykstorm/goldset](#)

- Three-runner AI eval framework (goldenDataset · llmJudge · structural assertions) shipped as a public GitHub Action.
- PR comment diffs the delta on every push; structural-assertion failures block merge.
- Published as \u{40}ykstorg/goldset v0.2.0 on npm and listed on GitHub Marketplace (Continuous Integration).

TypeScript esbuild OpenAI Anthropic GitHub Actions

Anchor [Creator](#) · [anchor-iota-ten.vercel.app/playground](#)

- Provenance-first RAG layer with cosine-floor refusal. Returns chunks when similarity >= 0.30; returns refused: true and zero chunks below floor.
- Live playground renders the retrieval decision — chunks above the floor, or an explicit refusal — for any query.
- Adaptive K per query intent, with a provenance array on every response so answers are traceable to source.

Postgres + pgvector Prisma 7 Next.js 15 OpenAI embeddings

Tripwire [Creator](#) · [github.com/ykstorm/tripwire](#)

- Mid-stream LLM safety. Token-by-token rule engine aborts the response stream on rule trip, before the bad token reaches the user.
- Sub-millisecond decision per chunk; zero buffering on happy path. Published as \u{40}ykstorg/tripwire on npm.

TypeScript Node async iterators streaming abort

Quickdraw [Creator](#) · [github.com/ykstorm/quickdraw](#)

- LLM streaming benchmark CLI. Captures TTFT, tokens-per-second, and dollar-per-1K tokens.
- Nightly run at 03:00 UTC across OpenAI + Anthropic; diffs vs. baseline and commits results to repo.
- Published as \u{40}ykstorg/quickdraw on npm.

TypeScript OpenAI SDK Anthropic SDK GHA cron

Stackup [Creator](#) · [github.com/ykstorm/stackup](#)

- Local Kubernetes lab spinning up in under 10 minutes via single make up target.
- ArgoCD + Argo Rollouts + Grafana + Prometheus on kind cluster — production-shaped stack for learning.

Kubernetes ArgoCD Helm kind Grafana

Codecraft [Creator](#) · [codecraft-ai-tau.vercel.app](#)

- Boots a real Next.js dev server inside the browser tab via WebContainers — no backend, no local install. The whole cross-origin-isolation (COOP/COEP) boundary is what makes SharedArrayBuffer and the in-tab runtime work.

WebContainers Monaco xterm Next.js COOP/COEP

FREELANCE WORK

Homesty.ai Solo Backend & AI Engineer · Remote · Mar 2025 – Present

- Sole engineer on a live buyer-side real-estate AI: Next.js 15, Postgres + pgvector, Prisma, GPT-4o and Claude streaming, behind rate limiting, intent routing, and Sentry.
- Built the chat path around refusal-first retrieval and a mid-stream guardrail — the AI declines when it has no grounded source rather than improvising, which in real estate is a liability, not a UX choice.
- Hardened those two patterns into standalone OSS the rest of the industry can use: Anchor (provenance RAG) and Tripwire (streaming safety).

Next.js 15 Postgres + pgvector Prisma 7 GPT-4o Claude Sentry Docker

EDUCATION

Manipal University Jaipur — B.Tech, Computer Science · Graduating 2026

Relevant coursework: Distributed Systems, Database Internals, Networks, ML Foundations

RECOGNITION

- Four npm packages under @ykstorsorg — Anvil published with npm build provenance (signed on GitHub Actions)
- Goldset listed on GitHub Marketplace (Continuous Integration)
- Seven open-source repositories, each with a working demo and green CI
- Portfolio with live npm + GitHub activity at lakshyaraj-dev.vercel.app

Latest version: github.com/ykstorm/resume · Apache 2.0